Tetrahedron 71 (2015) 9557–9563

Contents lists available at ScienceDirect

Tetrahedron

journal homepage: www.elsevier.com/locate/tet

Eight new alkaloids with PTP1B and α-glucosidase inhibitory activities from the medicinal mushroom *Hericium erinaceus*

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ARTICLE INFO

Article history:
Received 17 August 2015
Received in revised form 21 October 2015
Accepted 26 October 2015
Available online 27 October 2015

Keywords:
Hericium erinaceus
Alkaloids
PTP1B inhibition
α-Glucosidase inhibition

ABSTRACT

Eight new alkaloids (**1**–**8**), were isolated from the solid culture of *Hericium erinaceus*. The structures of new metabolites were established by spectroscopic methods. The absolute configuration of **1** was determined by electronic circular dichroism (ECD) calculation. Compounds **5**–**8** showed inhibitory activities against both protein tyrosine phosphatase-1B (PTP1B) and α-glucosidase. In the cytotoxicity assay, compounds **1**–**4** showed moderate cytotoxicity against K562 cells with IC_{50} values in the range of 11.4–18.2 μM, and also weak cytotoxicity against doxorubicin resistant K562 cells.

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1. Introduction

Hericium erinaceus is usually consumed as a popular edible and medicinal mushroom in East Asian.[1] It has been used as a herbal medicine or functional food for years. The extract of *H. erinaceus* has been reported to possess various pharmacological effects, including hypoglycemic, anticancer, and anti-inflammatory bioactivities.[2] Previous chemical investigations on *H. erinaceus* led to isolation of aromatic compounds,[3–5] diterpenes,[6–11] fatty acids,[12] sterols,[13,14] and polysaccharides.[15] Secondary metabolites from *H. erinaceus* exhibited various interesting bioactivities, such as cytotoxicity,[16] stimulatory activity for the nerve growth factor (NGF) synthesis,[9,10] stimulatory activity for the growth of lettuce,[17] peroxisome proliferators-activated receptor (PPAR) transactivational effect,[13] and inhibitory activity against α-glucosidase.[18] In our early research on *H. erinaceus*,[18] 10 new isoindolin-1-ones showing α-glucosidase inhibitory activity (ranging from 5.3 to 145.1 μM) were isolated from the solid culture of this mushroom. To obtain the minor alkaloids containing in the culture extract, we fermented this fungus in a larger scale and made a further chemical study. As a result, eight new alkaloids, erinacerins M-T (**1**–**8**) were isolated and identified. Herein, we described the isolation, structure determination, and biological activities of **1**–**8**.

2. Results and discussion

The ethyl acetate extract of *H. erinaceus* fermented on rice was fractionated by silica gel column chromatography (CC). The fractions containing minor alkaloids were further separated by silica gel and Sephadex LH-20 CC, and reversed-phase high-performance liquid chromatography to afford eight new alkaloids, erinacerins M-T (**1**–**8**).

Erinacerin M (**1**) was obtained as yellow oil. A molecular formula of $C_{13}H_{15}NO_4$ (seven unsaturation degrees) was deduced for **1** on the basis of its HRTOFMS at *m/z* $[M+H]^+$ 250.1080. The ^1H, ^{13}C NMR, and HSQC spectra of **1** showed resonances due to the presence of three methyls [δ_H 0.86 (3H, d, *J*=6.5 Hz), 1.02 (3H, d, *J*=6.5 Hz), 2.53 (s); δ_C 18.7, 18.8, and 31.0], one oxygenated methylene [δ_H 5.31 (1H, d, *J*=15.5 Hz), 5.56 (1H, d, *J*=15.5 Hz); δ_C 64.8], two methines, two pairs of olefinic carbons [δ_H/δ_C 6.49 (s)/118.6, 8.32 (s)/145.4; δ_C 124.9, 141.1], and three carbonyl moieties [δ_C165.5, 175.5, 196.9] (Table 1). The ^1H–^1H COSY correlations of H_3-13(H_3-14)–H-12–H-7, as well as the HMBC correlations from H_3-13(14) to C-12 and C-7,

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http://dx.doi.org/10.1016/j.tet.2015.10.068
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Table 1
^{1}H and ^{13}C NMR data of **1**–**4** (DMSO-d_6)

No.	**1**		**2**		**3**		**4**	
	δ_C	δ_H (m, *J* in Hz)	δ_C	δ_H (m, *J* in Hz)	δ_C	δ_H(m, *J* in Hz)	δ_C	δ_H (m, *J* in Hz)
1	145.4	8.32 s	144.1	8.50 s	144.1	8.50 s	143.6	8.35 s
2	124.9		124.0		124.0		124.0	
3	175.5		175.7		175.7		175.7	
4	118.6	6.49 s	120.3	6.43 s	120.4	6.42 s	120.4	6.38 s
5	141.1		151.4		151.4		151.2	
6								
7	67.5	4.84 d (10.5)	64.6	4.63 d (10.6)	65.7	4.57 d (10.6)	65.8	4.41 d (10.5)
8	165.5							
9	64.8	5.56 d (15.5); 5.31 d (15.5)	59.1	4.47 s	59.2	4.46 s	59.1	4.36 s
10	196.9		196.8		196.8		196.8	
11	31.0	2.53 s	31.0	2.54 s	31.0	2.53 s	31.1	2.51 s
12	29.4	2.36 m	37.1	2.07–2.09[a]	31.4	2.27 m	31.0	2.24 m
13	18.7	0.86 d (6.5)	24.1	1.17 m; 1.04 m	18.3	0.77 d (6.5)	18.0	0.74 d (6.5)
14	18.8	1.02 d (6.5)	10.6	0.81 t (7.0)	18.6	1.05 d (6.5)	18.7	1.01 d (6.5)
15			14.8	1.01 d (6.5)				
1′			168.2		168.1		168.2	
2′			57.5	4.12 dd (8.1, 5.8)	56.7	4.14 dd (7.9, 5.8)	53.4	4.47 ddd (9.8, 8.3, 4.8)
3′			29.7	2.09[a]	36.1	1.79 m	36.5	3.09 dd (13.8, 4.8); 2.84 dd (13.8, 9.8)
4′			19.0	0.80 d (6.5)	24.6	1.36 m; 1.15 m	137.0	
5′			17.8	0.87 d (6.5)	11.3	0.79 t (7.0)	128.0	7.09–7.14[b]
6′			172.3		15.5	0.80 d (6.7)	129.0	7.09–7.14[b]
7′					172.3		126.4	7.09–7.14[b]
8′							129.0	7.09–7.14[b]
9′							128.0	7.09–7.14[b]
10′							172.3	
9-OH				5.85 br s		5.86 br s		
2′-OH				8.68 d (8.1)		8.69 d (7.9)		8.94 d (8.3)
COOH				12.79 br s		12.77 br s		

[a,b]Signals overlapped with each other.

from H-12 to C-7 and C-8 assigned the substructure of an isobutanoic acid moiety. HMBC correlations of from H-1 to C-5, C-2, and C-4, H-4 to C-5, C-9, and C-10, from H_3-11 to C-2 and C-10, as well as the NMR data comparison between one and aspernigrin A confirmed the presence of 2-acetyl-5-(hydroxymethyl)pyridin-4(1H)-one moiety.[19] Furthermore, the HMBC correlations from H-7 to C-1 and C-5, from H_2-9 to C-8 established the planar structure of **1** as described in Fig. 1 (Fig. 2). To determine the absolute configuration at C-7, an electronic circular dichroism (ECD) calculation method was applied. A conformational analysis of one isomer of **1a** (7*S*) afforded nine conformers optimized with Gaussian09 software package at the B3LYP/6-31G(d) basis set level by density-functional theory (DFT). As shown in Fig. 3, the calculated ECD curve of **1a** by Boltzmann-weighting of the conformers was superimposable with the experimental CD, indicating the 7*S* absolute configuration of **1** (Fig. 4).

The molecular formula of erinacerin N (**2**) was established as $C_{19}H_{28}N_2O_6$ on the basis of its HRTOFMS (*m/z* $[M+H]^+$ 381.2026) and NMR data (Table 1). A NMR data comparison between **1** and **2** confirmed the presence of a 2-acetyl-5-(hydroxymethyl)pyridin-

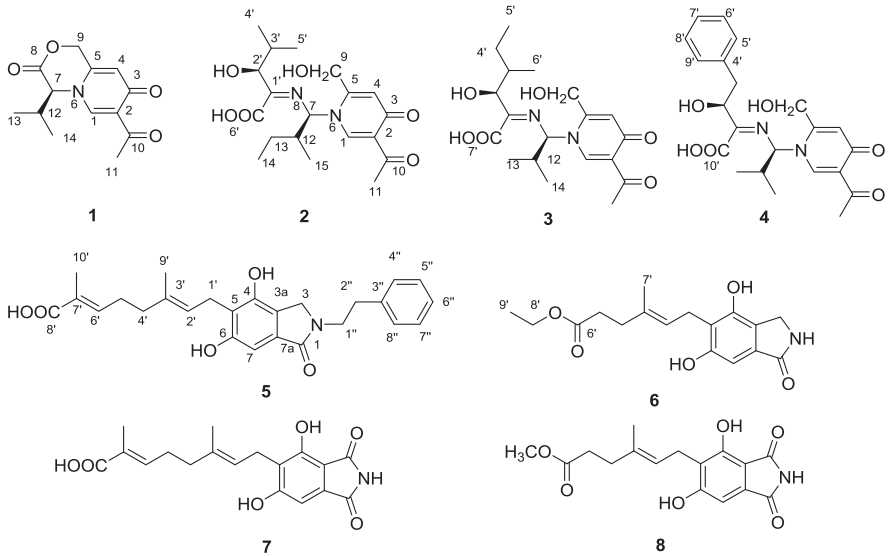


Fig. 1. Structures of compounds **1**–**8**.

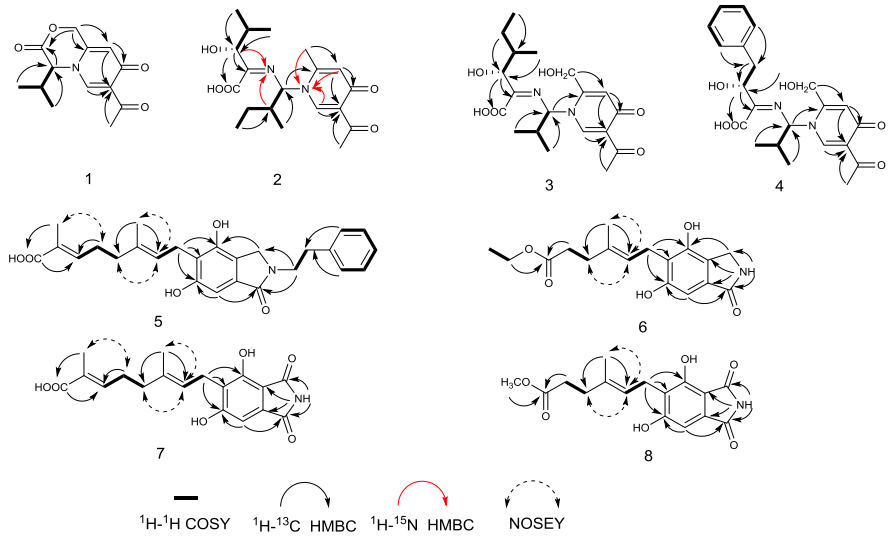


Fig. 2. Key ^{1}H–^{1}H COSY, HMBC, and NOSEY correlations of **1**–**8**.

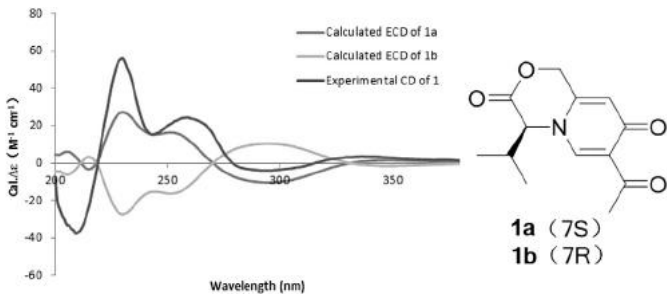


Fig. 3. Calculated and experimental CD spectra of **1**.

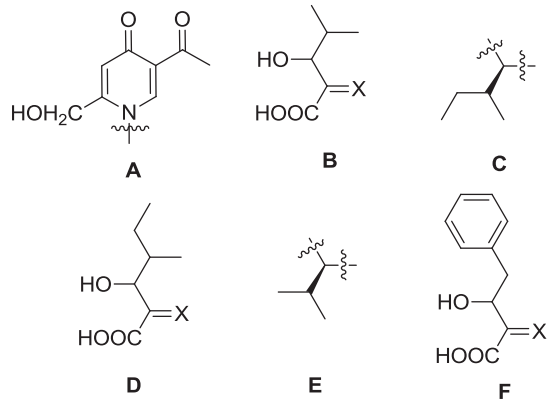


Fig. 4. Structural moieties of A–F.

4(1H)-one moiety (fragment A) in **2**. The HMBC correlations of H_3-4′ (H_3-5′)/C-2′, C-3′, H-2′/C-1′ (δ_C 168.2), C-6′ (δ_C 172.3), HO-2′/C-1′, C-2′, and C-3′, as well as the ^{1}H–^{1}H COSY correlations of H_3-4′ (H_3-5′)–H-3′–H-2′ confirmed the presence of the structure of fragment B. Similarly, the ^{1}H–^{1}H COSY correlations of H_3-14–H_2-13–H-12–H-7, H_3-15–H-12, together with the HMBC correlations from H_3-14 to C-12 and C-13, from H_3-15 to C-7, C-12, and C-13 supported the structure of fragment C. The HMBC correlation from H-7 to C-1 and C-4 connected the fragment A with C by the chemical bond between C-7 and N-6. To determine the linkage of the structural fragment B, the ^{1}H–^{15}N HMBC spectrum was measured. The resonances responsible for two nitrogen atoms at 122.4 and 149.2 were observed. The HMBC correlations from H-7 to C-1′, from H-1, H-4, and H_2-9 to the nitrogen atom at 149.2, and from H-2′ and H-12 to the nitrogen atom at 122.4 connected the structural fragment B with C by a chemical bond between C-7 and N-8. The CD spectrum of **2** showed positive Cotton effects at 230 and 263 nm, and negative Cotton effect at 290 nm, which is nearly identical with those of **1**, indicating the *R* absolute configuration at C-7. The absolute configuration at C-2′ was determined by the circular dichroism data on the basis of the circular dichroism of an in situ formed complex with [$Rh_2(OCOCF_3)_4$], with the inherent CD data subtracted. The sign of the E band (at ca. 350 nm) in the CD spectrum of the complex is correlated with the absolute configuration of a secondary alcohol.[20] The Rh complex of **2** displayed a positive E band (Fig. 5), correlating with 2*S* absolute configuration. The absolute configuration at C-12 was left unsolved from the current data.

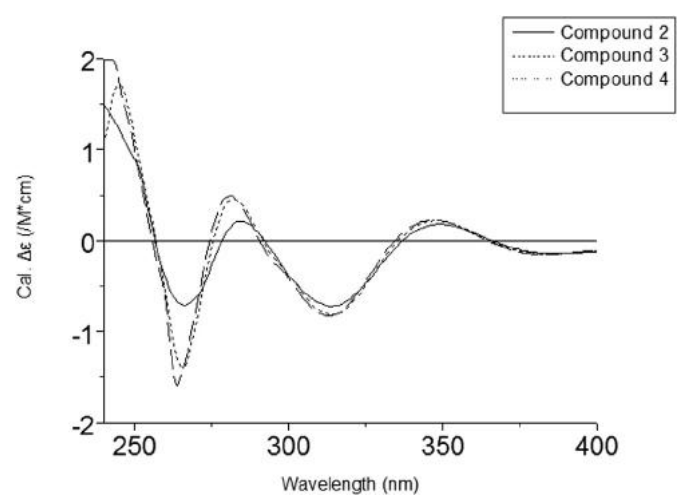


Fig. 5. CD data of the Rh complexes **2**–**4**.

The elemental composition of erinacerin O (**3**) was determined to be $C_{19}H_{28}N_2O_6$ by its HRTOFMS (*m/z* [M+H]$^+$ 381.2027) and NMR data (Table 1). The ^{1}H and ^{13}C spectra of **3** were similar with those of **2**, indicating the similar structural feature between these two compounds. Interpretation of ^{1}H–^{1}H COSY and HMBC spectra of **3**

revealed the presence of three structural fragments A, D, and E. The HMBC correlations from H-7 to C-1, C-5, and C-1′ confirmed the linkage among structural fragments. The CD spectrum of **3** resembled with those of **1** and **2**, suggesting the 7*R* absolute configuration in **3**. The Rh complex of **3** displayed a positive E band (Fig. 5), indicating the 2*S* absolute configuration.

Erinacerin P (**4**) was assigned the molecular formula of $C_{22}H_{26}N_2O_6$ on the basis of its HRTOFMS (*m/z* $[M+H]^+$ 415.1866). A comparison of NMR data between **4** and **3** indicated the presence of the structural moieties A and E, which was further supported by a detailed elucidation of its NMR. A monosubstituted aryl ring was deduced from the NMR data at δ_H 7.09–7.14 (5H) and δ_C 126.4, 128.0 (two carbons), 129.0 (two carbons), 126.4. The structural fragment F was confirmed by $^1H-^1H$ COSY correlations of H-2′–H-3′, and HMBC correlations of H-2′/C-1′, C-3′, C-4′, and C-10′, H-3′/C-1′, C-2′, C-4′, C-5′, ad C-9′. As described in **2** and **3**, the HMBC correlations from H-7 to C-1, C-5, and C-1′ established the planar structure of **4**. The absolute configuration of **4** was determined as 2*S* and 7*R* by the comparison of its CD data with those of **2**, and the CD data of its Rh complex.

The molecular formula of erinacerin Q (**5**) was determined to be $C_{26}H_{29}NO_5$ (13 degrees of unsaturation) on the basis its HRTOFMS (*m/z* $[M+H]^+$ 436.2125) and NMR data (Table 2). The ^1H and ^{13}C NMR spectroscopic data of **5** were very similar to the know compound (E)-5-(3,7-dimethylocta-2,6-dien-1-yl)-4-hydroxy-6-methoxy-2-phenethylisoindolin-1-one,[21] except for the loss of a methyl group and the presence of an additional carbonyl moiety at δ_C 168.8. HMBC correlations from H_3-10′ to C-6′, C-7′, and C-8′, and from H-6′ to C-4′, C-5′, C-7′, C-8′, and C-10′ confirmed the structural variation at C-8′ in **5**. NOE correlations between H-2′ and H_2-4′ and between H_2-5′ and H_3-10′ confirmed the *E* configuration for two double bonds. Based on above analysis, the structure of five was proposed.

Erinacerin R (**6**) was assigned a molecular formula of $C_{17}H_{21}NO_5$ (eight degrees of unsaturation) on the basis its HRTOFMS (*m/z* $[M+H]^+$ 320.1498) and NMR data (Table 2). The ^1H and ^{13}C NMR data of **6** were nearly identical to those of erinacerin G,[18] except for the presence of an ethoxyl group [δ_H 1.05 (3H, t, *J*=7.0), 3.44 (2H, q, *J*=7.0); δ_C 18.6, 56.1] instead of the methoxyl moiety at C-6′ in erinacerin G. The $^1H-^1H$ COSY correlations of H_2-8′–H_3-9′, and HMBC correlations from H_3-9′ to C-8′, from H_2-8′ to C-6′ and C-9′ indicated the substitution of an ethoxyl group at C-6′. The *E* configuration for the double bond between C-2′ and C-3′ was determined by the NOE correlation between H-2′ and H_2-4′.

The elemental composition of erinacerin S (**7**) was determined to be $C_{18}H_{19}NO_6$ by analysis of its HRTOFMS (*m/z* $[M+H]^+$ 346.1292) and NMR data (Table 2). The ^1H, ^{13}C, and HSQC spectra of **7** resembled with those of erinacerin C, except for the absence of the methylene at C-3 (δ_H 4.13) in erinacerin C and the presence of one additional carbonyl moiety at δ_C 169.0 in the ^{13}C NMR spectrum of **7**. The HMBC correlations from HN (δ_H 10.78) to C-1 (δ_C 168.8), C-3 (δ_C 169.0), C-3a, and C-7a confirmed the presence of an isoindoline-1, 3-dione moiety in **7**. NOE correlations of H-2′/H_2-4′, and H_2-5′/H_3-10′ determined the *E* configuration for two double bonds.

Erinacerin T (**8**) was assigned a molecular formula of $C_{16}H_{17}NO_6$ by its HRTOFMS (*m/z* $[M+H]^+$ 320.1135) and NMR data (Table 2). The ^1H, ^{13}C, and HSQC NMR data of **8** showed resonances ascribable for one methyl group (δ_H 1.71, s; δ_C 15.9), three methylenes, one methoxyl group (δ_H 3.51, s; δ_C 51.1), three exchangeable protons (δ_H 9.57, s; 10.69, s; 10.78, s), one trisubstituted olefin [δ_H 5.14, t, *J*=7.1 Hz; δ_C 122.0 (=CH), 132.4], a pentasubstituted benzene (δ_H 6.74, s; δ_C 102.7, 107.9, 121.1, 133.5, 153.4, and 161.7), and three carbonyl groups (δ_C 169.0, 169.5, and 173.0). The ^1H and ^{13}C spectral data of **8** were very similar to those of erinacerin G, except for the loss of one methylene group at C-3 in erinacerin G and the presence

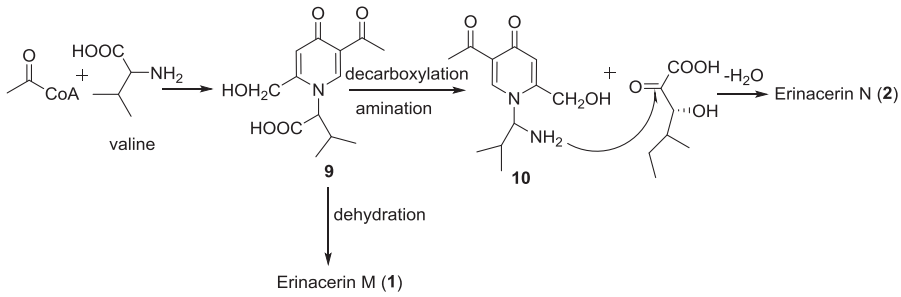
Table 2
^1H and ^{13}C NMR data of **5**–**8** (DMSO-d_6)

No.	**5**		**6**		**7**		**8**	
	δ_C	δ_H (m, *J* in Hz)	δ_C	δ_H (m, *J* in Hz)	δ_C	δ_H(m, *J* in Hz)	δ_C	δ_H (m, *J* in Hz)
1	167.8		170.5		168.8		169.0	
3	43.2	4.16 s	43.0	4.14 s	169.0		169.5	
3a	131.1		131.4		134.2		133.5	
4	150.1		150.5		153.4		153.4	
5	118.6		118.6		121.1		121.1	
6	156.2		156.2		161.7		161.7	
7	100.4	6.62 s	100.3	6.63 s	102.8	6.75 s	102.7	6.74 s
7a	118.7		121.1		107.9		107.9	
1′	22.4	3.28 d (7.0)	22.5	3.29 d (7.1)	22.0	3.27 d (7.0)	21.9	3.25 d (7.1)
2′	123.0	5.21 t (7.0)	122.6	5.20 t (7.1)	121.9	5.16 t (7.0)	122.0	5.14 t (7.1)
3′	133.2		132.9		132.3		132.4	
4′	37.9	2.01 t (7.5)	34.4	2.14 t (7.6)	37.8	2.02 t (7.5)	34.2	2.16 t (7.6)
5′	26.9	2.20 m	32.7	2.24 t (7.6)	26.8	2.20 m	32.3	2.34 t (7.6)
6′	141.3	6.61 t (7.5)	174.3		141.1	6.59 t (7.5)	173.0	
7′	127.7		16.0	1.73 s	127.7		15.9	1.71 s
8′	168.8		56.1	3.44 q (7.0)	169.4		51.1	3.51 s
9′	16.0	1.74 s	18.6	1.05 t (7.0)	16.0	1.74 s		
10′	12.2	1.71 s			12.2	1.70 s		
1″	47.9	3.71 t (7.2)						
2″	34.1	2.88 t (7.2)						
3″	139.1							
4″	128.4	7.18–7.23[a]						
5″	128.6	7.18–7.23[a]						
6″	126.3	7.28 t (7.6)						
7″	128.6	7.21[a]						
8″	128.4	7.21[a]						
4-OH		9.22 s				10.78 s		10.78 s
6-OH		9.52 s		9.51 s		9.55 s		9.57 s
NH				8.28 s		10.78 s		10.69 s

[a] Signals overlapped with each other.

of an additional carbonyl moiety. Analysis of $^1H-^1H$ COSY, HMBC, and ROESY spectra of **8** confirmed its structure as described in Fig. 1.

The biosynthetic pathway for compounds **1** and **2** was postulated (Scheme 1). The key intermediate product **9** was synthesized from four molecules of acetyl CoA and one molecule of valine. Compound **1** was derived from **9** by dehydration. Compound **10** obtained from **9** by decarboxylation and amination reactions was further condensed with 3-hydroxy-4-methyl-2-oxohexanoic acid to form compound **2**. Compounds **3** and **4** were produced by the similar way as that of **2**.



Scheme 1. The biosynthetic pathway of **1** and **2**.

In our early research, the alkaloids isolated from *H. erinaceus* were found to have good inhibitory activity against α-glucosidase. In the current study, we used protein tyrosine phosphatase-1B (PTP1B) and α-glucosidase inhibition assays to evaluate the hypoglycemic activity of new compounds **1**–**8**. Compounds **5**–**8** showed inhibitory activity against both PTP1B phosphatase and α-glucosidase (Table 3). The in vitro cytotoxicity of **1**–**8** was evaluated against wild K562 and adriamycin resistant K562 cell lines (K562/A). Compounds **1**–**4** showed moderate cytotoxicity against wild K562 cells with the IC_{50} values of 16.3, 18.2, 15.9, and 11.4 μM, respectively (Table 3). They also presented weak cytotoxicity against doxorubicin resistant K562 cells.

Table 3
PTP1B inhibitory, α-Glucosidase inhibitory and cytotoxic activities of **1**–**8**

	PTP1B inhibition (IC_{50}, μM)	α-glucosidase inhibition (IC_{50}, μM)	K562 (IC_{50}, μM)	K562/A (IC_{50}, μM)
1	>50	>50	16.3±2.1	98.7±9.8
2	>50	>50	18.2±1.9	156.3±14.3
3	>50	>50	15.9±1.9	162.7±12.3
4	>50	>50	11.4±1.1	128.9±19.1
5	29.1±4.3	12.7±2.1	>200	>200
6	42.1±5.1	23.3±3.1	71.0±12.1	174.3±23.1
7	28.5±3.4	19.5±3.2	>200	>200
8	24.9±6.1	20.1±4.1	29.7±2.3	>200
Positive control	Sodium vanadate 1.2±0.3	Acarbose 273.1±23.1	Adriamycin 0.6±0.1	Adriamycin 62.5±6.2

In summary, further chemical investigation on the culture extract of *H. erinaceus* afforded eight new alkaloids. Bioactivity evaluation confirmed the cytotoxicity of compounds **1**–**4** on K562 cells, and the inhibitory effect of **5**–**8** on PTP1B and α-glucosidase. The discovery of new alkaloids with hypoglycemic effect from the edible and medicinal mushroom *H. erinaceus* supports the usage of this mushroom as supplementary food or medicine for the prevention or treatment of diabetes mellitus type 2.

3. Experimental section

3.1. General experimental procedures

UV and IR spectral data were recorded on Thermo Genesys-10S UV–Vis and Nicolet IS5 FTIR spectrophotometer, respectively. Optical rotations and CD spectra were measured on a Perkin–Elmer 241 Polarimeter and a Jasco J-815 sepctropolarimeter, respectively. NMR spectral data were obtained with Bruker Avance-500 spectrometer (DMSO-d_6, δ_H 2.50/δ_C 40.0). HRTOFMS data were determined using an Agilent Accurate-Mass-Q-TOF LC/MS 6520 instrument. TLC was carried out on Silica gel $HSGF_{254}$ and the spots were detected by 10% H_2SO_4 and heating. Silica gel (Qingdao Haiyang Chemical Co., Ltd., People's Republic of China) and Sephadex LH-20 (Amersham Biosciences) were used for column chromatography. HPLC separation was performed on a Waters 1525 HPLC system equipped with UV/Vis 2489 detector and an ODS column (C_{18}, 250×9.4 mm, YMC Pak, 5 μm; UV detector: 220 nm) with a flow rate of 2.5 mL/min.

3.2. Fungal material

The strain used in this work was isolated from the fruiting body of *H. erinaceus* and identified by DNA sequences of the ITS1-5.8S-ITS2 regions of the rRNA gene. According to the previously reported method, the larger scale-up fermentation was carried out in sixty 500 mL Fernbach culture flasks each containing 80 g of rice and 120 mL of distilled water at 25 °C for 40 days.

3.3. Extraction and isolation

The fermented rice substrate was extracted repeatedly with EtOAc (3×10 L), and the organic solvent was evaporated under vacuum to give the crude extract (30.6 g). The crude extract was subsequently separated on a silica gel column (500 g of silica gel) eluted with hexane–ethyl acetate in a gradient elution (v/v, 100:0, 100:2, 100:5, 100:8, 100:10, 100:15, 100:20, 100:40, 100:50, each 2000 mL), followed by dichloromethane–methanol (v/v, 100:1, 100:2, 100:3, 100:5, 100:10, 100:15, 100:20, 0:100, each 2000 mL) to afford 20 fractions (HE-1–HE-20). Fractions seven and eight containing different secondary metabolites from our early report were selected for further investigation.

Fraction 7 (1.56 g) eluted with dichloromethane–methanol (v/v, 100:2) was further separated by Sephadex LH-20 column

chromatography (CC) using 50% methanol in water to give six subfractions (HE-7-1–HE-7-6). Compounds **1** (10.2 mg, t_R 20.3 min) and **4** (7.3 mg, t_R 22.3 min) was obtained from the subfraction HE-7-5 (70.2 mg) by RP-HPLC using 36% acetonitrile in water. Compounds **2** (21.3 mg) and **3** (17.9 mg) were obtained from the subfraction HE-7-3 (241 mg) by Sephadex LH-20 CC eluted with 52% methanol in water. Fraction 8 (1.56 g) eluted with dichloromethane–methanol (v/v, 100:10) was fractionated by silica gel CC with a $CHCl_3$–MeOH gradient elution (100:0–0:100) to yield 10 subfractions (GL-8-1–GL-8-10). Compounds **5** (12.2 mg, t_R 19.3 min), **6** (7.1 mg, t_R 23.1 min), **7** (4.5 mg, t_R 23.5 min), **8** (10.1 mg, t_R 25.2 min) were purified from fraction GL-8-7 (211.0 mg) by RP-HPLC using 40% methanol in water.

3.4. Spectral data of the new compounds

*3.4.1. Erinacerin M (**1**).* Yellow oil; $[\alpha]^{25}_D$ −10.21 (*c* 0.1, MeOH); UV (methanol) λ_{max} (log ε) 220 (4.50), 260 (4.25), 309 (2.10) nm; IR (neat) ν_{max} 3330, 2830, 2820, 1684, 1660, 1451, 1380, 1050, 800, 755, 620 cm^{-1}; for ^1H and ^{13}C NMR data see Table 1; positive HRTOFMS *m/z* $[M+H]^+$ 250.1080 (calcd for $C_{13}H_{16}NO_4$, 250.1079).

*3.4.2. Erinacerin N (**2**).* Yellow oil; $[\alpha]^{25}_D$ −12.50 (*c* 0.1, MeOH); UV (methanol) λ_{max} (log ε) 215 (4.63), 265 (4.50), 310 (2.10) nm; IR (neat) ν_{max} 3310, 2879, 2855, 1680, 1630, 1450, 1390, 1060, 824, 740, 625 cm^{-1}; for ^1H and ^{13}C NMR data see Table 1; positive HRTOFMS *m/z* $[M+H]^+$ 381.2026 (calcd for $C_{19}H_{29}N_2O_6$, 381.2026).

*3.4.3. Erinacerin O (**3**).* Yellow oil; $[\alpha]^{25}_D$ −15.20 (*c* 0.1, MeOH); UV (methanol) λ_{max} (log ε) 212 (4.23), 265 (4.20), 310 (2.10) nm; IR (neat) ν_{max} 3280, 2850, 2830, 1685, 1650, 1450, 1390, 1060, 824, 740, 624 cm^{-1}; for ^1H and ^{13}C NMR data see Table 1; positive HRTOFMS *m/z* $[M+H]^+$ 381.2027 (calcd for $C_{19}H_{29}N_2O_6$, 381.2026).

*3.4.4. Erinacerin P (**4**).* Yellow oil; $[\alpha]^{25}_D$ −40.00 (*c* 0.1, MeOH); UV (methanol) λ_{max} (log ε) 218 (4.50), 260 (3.10), 307 (2.10) nm; IR (neat) ν_{max} 3328, 2870, 2852, 1683, 1670, 1660, 1451, 1420, 1400, 1050, 800, 755, 620 cm^{-1}; for ^1H and ^{13}C NMR data see Table 1; positive HRTOFMS *m/z* $[M+H]^+$ 415.1866 (calcd for $C_{22}H_{27}N_2O_6$, 415.1869).

*3.4.5. Erinacerin Q (**5**).* Colourless powder; UV (methanol) λ_{max} (log ε) 220 (4.35), 270 (3.20), 310 (2.00) nm; IR (neat) ν_{max} 3310, 2850, 2845, 1670, 1645, 1600, 1451, 1431, 1400, 1031, 625 cm^{-1}; for ^1H and ^{13}C NMR data see Table 2; positive HRTOFMS *m/z* $[M+H]^+$ 436.2125 (calcd for $C_{26}H_{30}NO_5$, 436.2124).

*3.4.6. Erinacerin R (**6**).* Yellow powder; UV (methanol) λ_{max} (log ε) 220 (4.35), 270 (3.20), 310 (2.00) nm; IR (neat) ν_{max} 3302, 2842, 2800, 1670, 1660, 1645, 1425, 1400, 1025, 700, 600 cm^{-1}; for ^1H and ^{13}C NMR data see Table 2; positive HRTOFMS *m/z* $[M+H]^+$ 320.1498 (calcd for $C_{17}H_{22}NO_5$, 320.1498).

*3.4.7. Erinacerin S (**7**).* Yellow powder; UV (methanol) λ_{max} (log ε) 213 (4.30), 275 (3.20), 310 (2.00) nm; IR (neat) ν_{max} 3310, 2843, 2801, 1675, 1650, 1640, 1400, 1380, 1025, 723, 600 cm^{-1}; for ^1H and ^{13}C NMR data see Table 2; positive HRTOFMS *m/z* $[M+H]^+$ 346.1291 (calcd for $C_{18}H_{20}NO_6$, 346.1291).

*3.4.8. Erinacerin T (**8**).* Yellow oil; UV (methanol) λ_{max} (log ε) 220 (4.30), 280 (3.15), 310 (2.00) nm; IR (neat) ν_{max} 3305, 2840, 2812, 1680, 1640, 1630, 1423, 1400, 1380, 1023, 723, 623 cm^{-1}; for ^1H and ^{13}C NMR data see Table 2; positive HRTOFMS *m/z* $[M+H]^+$ 320.1135 (calcd for $C_{16}H_{18}NO_6$, 320.1134).

3.5. Computational details[22]

A pair of enantiomers (**1a** and **1b**) was proposed for computation. The method and software using for ECD calculation in this work was the same as that previously reported.[20] Systematic conformational analysis of **1a** afforded nine conformers that was optimized with Gaussian09 software package at the B3LYP/6-31G(d) basis set level using density-functional theory (DFT). ECD spectra were stimulated using a Gaussian function with a half-bandwidth of 0.4 eV. The overall ECD spectra were attained on the basis of Boltzmann weighting of each conformer.

3.6. Absolute configuration of the C-2′ in 2[24]

A sample of **2** (1 mg) was dissolved in a dry solution of $CHCl_3$ (0.5 mL). The first CD spectrum was recorded. Then, the $CHCl_3$ solution of **2** was mixed with $[Rh_2(OCOCF_3)_4]$ complex (1.1 mg), and its time evolution was monitored until stationary (ca. 5 min after mixing). The inherent CD was subtracted. The observed sign of the E band at ca. 350 nm in the induced CD spectrum was correlated to the absolute configuration of the C-2′ moiety.

3.7. α-Glucosidase inhibitory assay

As described in our early work,[23] the bioassay was conducted using a 96-well plate, and the absorbance was determined at 405 nm using a Spectra max 190 microplate reader (Molecular Devices Inc.). The control was prepared by adding phosphate buffer instead of the sample in the same way as the test. The blank was prepared by adding phosphate buffer instead of the α-glucosidase. The inhibition rates (%)=$[(OD_{control}-OD_{control\ blank})-(OD_{test}-OD_{test\ blank})]/(OD_{control}-OD_{control\ blank})\times 100\%$. Acarbose was utilized as the positive control with IC_{50} of 273.1 μM.

3.8. PTP1B inhibitory assay

To each of 96-wells (final volume: 100 μL) in a microtiter plate was added 25 μL of PTP1B (0.05 μg, human recombinant, Plymouth Meeting), 175 μL of phosphate buffer (50 mM, pH 6.8) and 2 mM *p*-nitrophenyl phosphate (*p*-NPP) in a buffer containing 50 mM citrate (pH 6.0), 0.1 M NaCl, 1 mM EDTA, and 1 mM dithiothreitol (DTT), with or without the test compounds. The reaction was quenched with 10 M NaOH after incubation at 37 °C for 30 min. The produced *p*-nitrophenol was measured using the absorbance at 405 nm. The non-enzymatic hydrolysis of 2 mM *p*-NPP was adjusted by measuring the increased absorbance at 405 nm acquired in the absence of PTP1B enzyme. The IC_{50} values of the compounds were determined utilizing linear regression analysis. Sodium vanadate was utilized as the positive control with IC_{50} of 1.2±0.3 μM.

3.9. Cytotoxicity assay

The cytotoxicity against K562 and K563A cell lines of compounds **1**–**8** were tested using the MTT method as previously reported.[22]

3.10. Statistical analysis

The bioactivity values were expressed as means of three independent experiments, and each carried out in triplicate.

Acknowledgements

Financial supports of the Ministry of Science and Technology of China (2014CB138304) and the National Nature Science Foundation (21472233) are acknowledged. Dr. Jinwei Ren and Dr. Wenzhao Wang (State Key Laboratory of Mycology, Institute of Microbiology, Chinese Academy of Sciences) are appreciated for their help in measuring the NMR and MS data. Dr. Jonathan Hunt (State University of New York) are appreciated for help in checking the grammar and language.

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